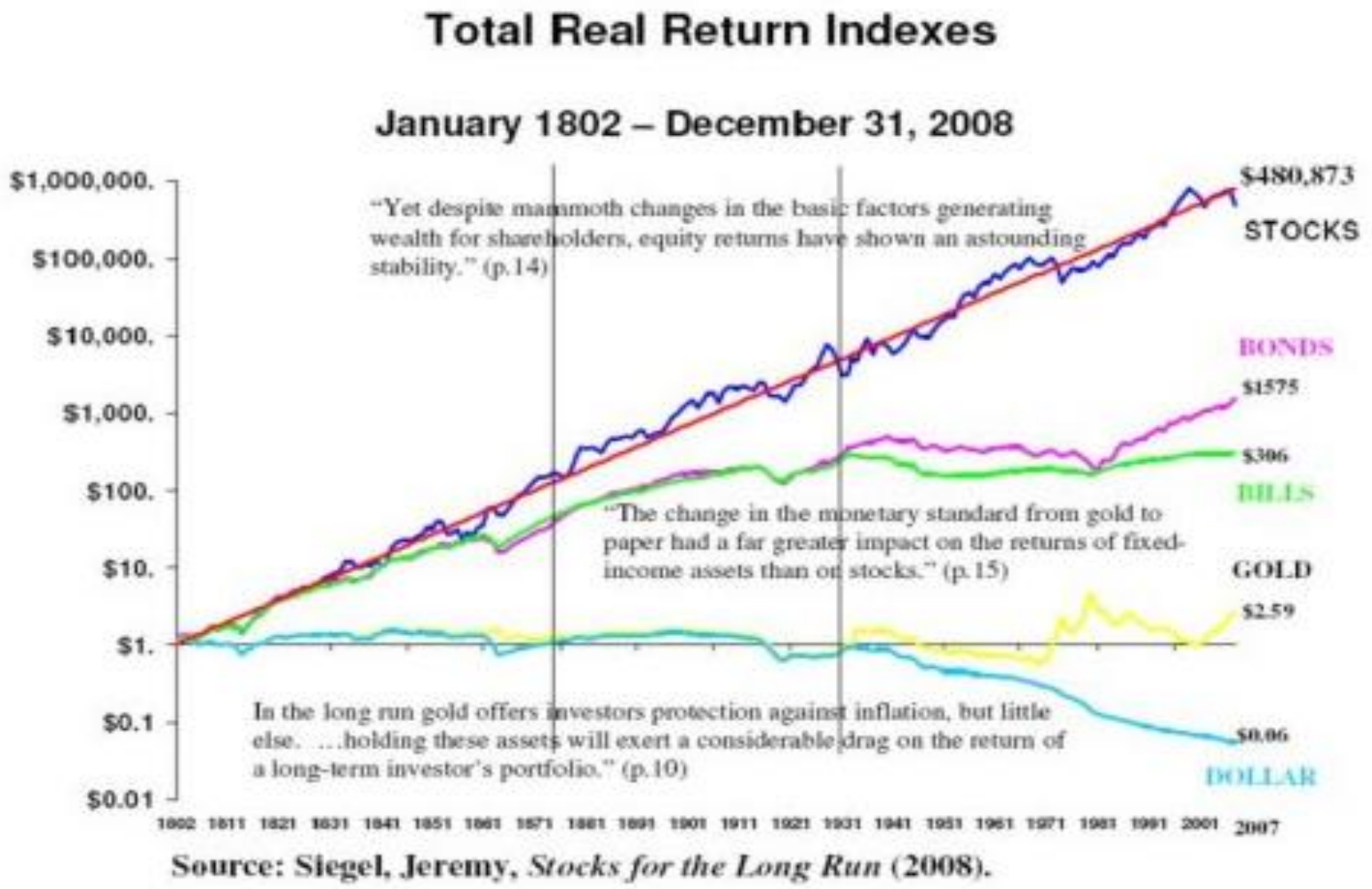
Valuation: Lecture Note Packet 1
Intrinsic Valuation

Prof. Jinfan Zhang
CUHK, Shenzhen SME


Total Real Return Indexes
January 1802 – December 31, 2008
$1,000,000.
$100,000.
$10,000.
$1,000.
$100.
$10.
$1.
$0.1
$0.01
"Yet despite mammoth changes in the basic factors generating wealth for shareholders, equity returns have shown an astounding stability." (p.14)
"The change in the monetary standard from gold to paper had a far greater impact on the returns of fixed-income assets than on stocks." (p.15)
In the long run gold offers investors protection against inflation, but little else. …holding these assets will exert a considerable drag on the return of a long-term investor's portfolio." (p.10)
$480,873
STOCKS
BONDS
$1575
$306
BILLS
GOLD
$2.59
$0.06
DOLLAR
1802 1811 1821 1831 1841 1851 1861 1871 1881 1891 1901 1911 1921 1931 1941 1951 1961 1971 1981 1991 2001 2007
Source: Siegel, Jeremy, Stocks for the Long Run (2008).

The essence of intrinsic value

- In intrinsic valuation, you value an asset based upon its fundamentals (or intrinsic characteristics).
- For cash flow generating assets, the intrinsic value will be a function of the magnitude of the expected cash flows on the asset over its lifetime and the uncertainty about receiving those cash flows.
- Discounted cash flow valuation is a tool for estimating intrinsic value, where the expected value of an asset is written as the present value of the expected cash flows on the asset, with either the cash flows or the discount rate adjusted to reflect the risk.

The two faces of discounted cash flow valuation

- The value of a risky asset can be estimated by discounting the expected cash flows on the asset over its life at a risk-adjusted discount rate:

$$\text{Value of asset} = \frac{E(CF_1)}{(1+r)} + \frac{E(CF_2)}{(1+r)^2} + \frac{E(CF_3)}{(1+r)^3} \ldots\ldots + \frac{E(CF_n)}{(1+r)^n}$$

where the asset has an n-year life, $E(CF_t)$ is the expected cash flow in period t and r is a discount rate that reflects the risk of the cash flows.

- Alternatively, we can replace the expected cash flows with the guaranteed cash flows we would have accepted as an alternative (certainty equivalents) and discount these at the riskfree rate:

$$\text{Value of asset} = \frac{CE(CF_1)}{(1+r_f)} + \frac{CE(CF_2)}{(1+r_f)^2} + \frac{CE(CF_3)}{(1+r_f)^3} \ldots\ldots + \frac{CE(CF_n)}{(1+r_f)^n}$$

where CE(CFt) is the certainty equivalent of $E(CF_t)$ and r_f is the riskfree rate.

Risk Adjusted Value: Two Basic Propositions

- The value of an asset is the risk-adjusted present value of the cash flows:

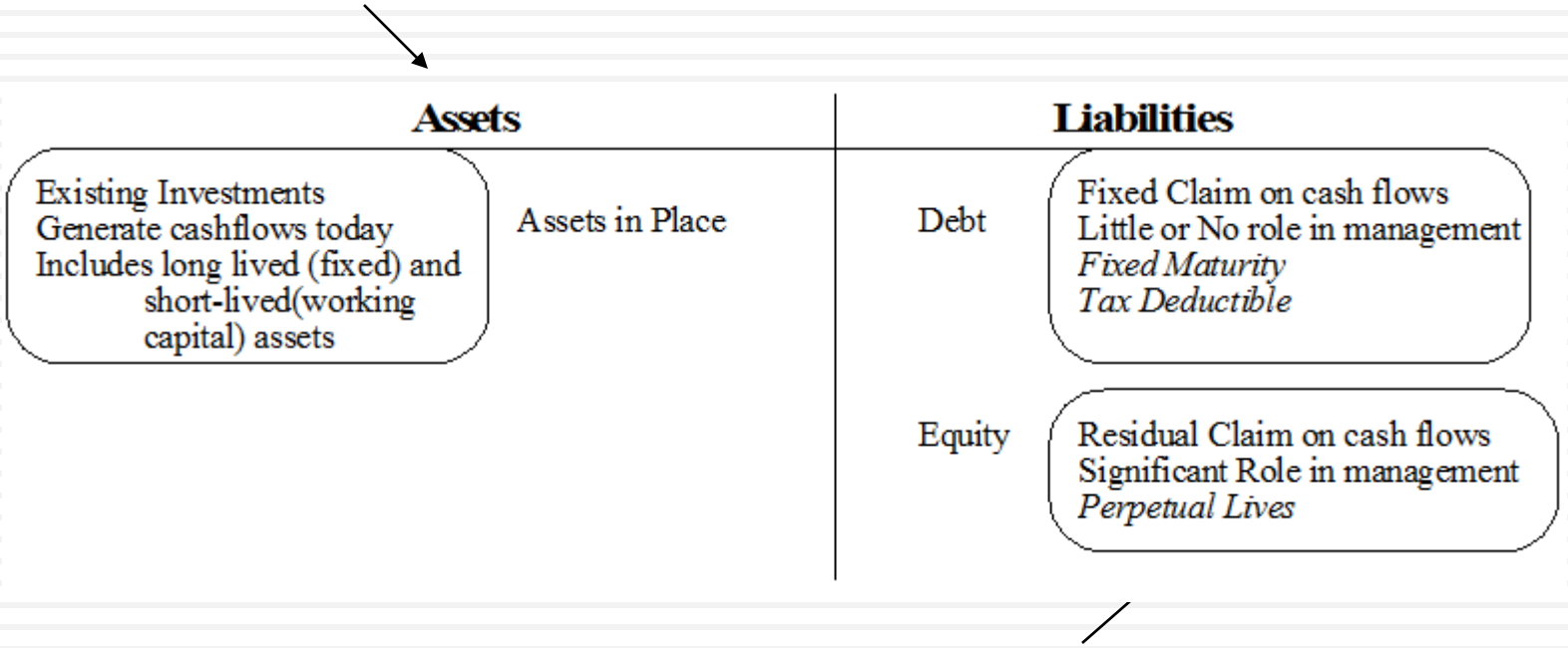
$$\text{Value of asset} = \frac{E(CF_1)}{(1+r)} + \frac{E(CF_2)}{(1+r)^2} + \frac{E(CF_3)}{(1+r)^3} \ldots\ldots + \frac{E(CF_n)}{(1+r)^n}$$

$$\text{Value of asset} = \frac{CE(CF_1)}{(1+r_f)} + \frac{CE(CF_2)}{(1+r_f)^2} + \frac{CE(CF_3)}{(1+r_f)^3} \ldots\ldots + \frac{CE(CF_n)}{(1+r_f)^n}$$

1. If IT does not affect the expected cash flows or the riskiness of the cash flows, IT cannot affect value.
2. For an asset to have value, the expected cash flows have to be positive some time over the life of the asset.
3. Assets that generate cash flows early in their life will be worth more than assets that generate cash flows later; the latter may however have greater growth and higher cash flows to compensate.

DCF Choices: Equity Valuation versus Firm Valuation

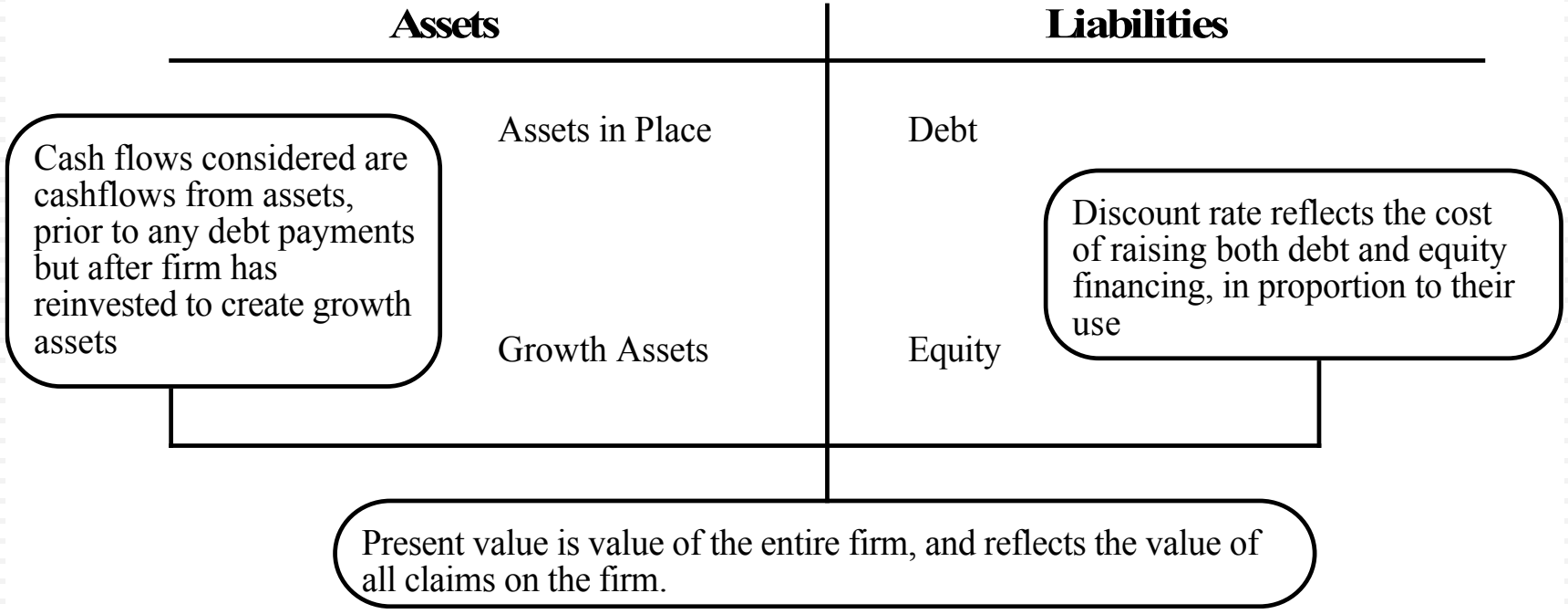
Firm Valuation: Value the entire business



Equity valuation: Value just the equity claim in the business

Firm Valuation

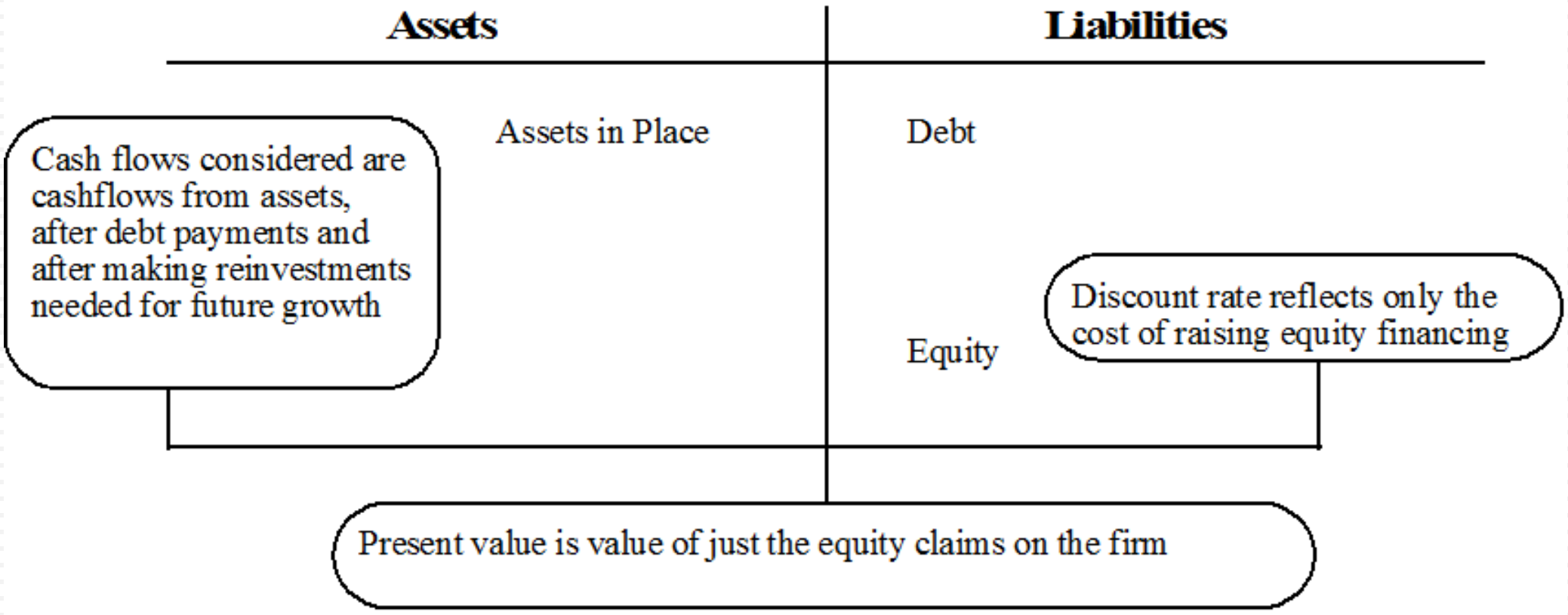
Figure 5.6: Firm Valuation



book value ⟺ market value → DCF

(history) (looking forward)

Equity Valuation



Cash Flows and Discount Rates

- Assume that you are analyzing a company with the following cash flows for the next five years.

Year	CF to Equity	Interest Exp	CF to Firm
1	$ 50	$ 40	$ 90
2	$ 60	$ 40	$ 100
3	$ 68	$ 40	$ 108
4	$ 76.2	$ 40	$ 116.2
5	$ 83.49	$ 40	$ 123.49
Terminal Value			
	$ 1603.0		$ 2363

- Assume also that the cost of equity is 13.625% and the firm can borrow long term at 5%. The current market value of equity is $1,073 and the value of debt outstanding is $800.

Equity versus Firm Valuation

- Method 1: Discount CF to Equity at Cost of Equity to get value of equity
  - Cost of Equity = 13.625%
  - Value of Equity = $50/1.13625 + 60/1.13625^2 + 68/1.13625^3 + 76.2/1.13625^4 + (83.49+1603)/1.13625^5$ = **$1073**
- Method 2: Discount CF to Firm at Cost of Capital to get value of firm
  - Cost of Debt = 5%

Cost of Capital = 13.625% (1073/1873) + 5% (800/1873) = 9.94%

- PV of Firm = $90/1.0994 + 100/1.0994^2 + 108/1.0994^3 + 116.2/1.0994^4 + (123.49+2363)/1.0994^5$ = $1873
- Value of Equity = Value of Firm - Market Value of Debt

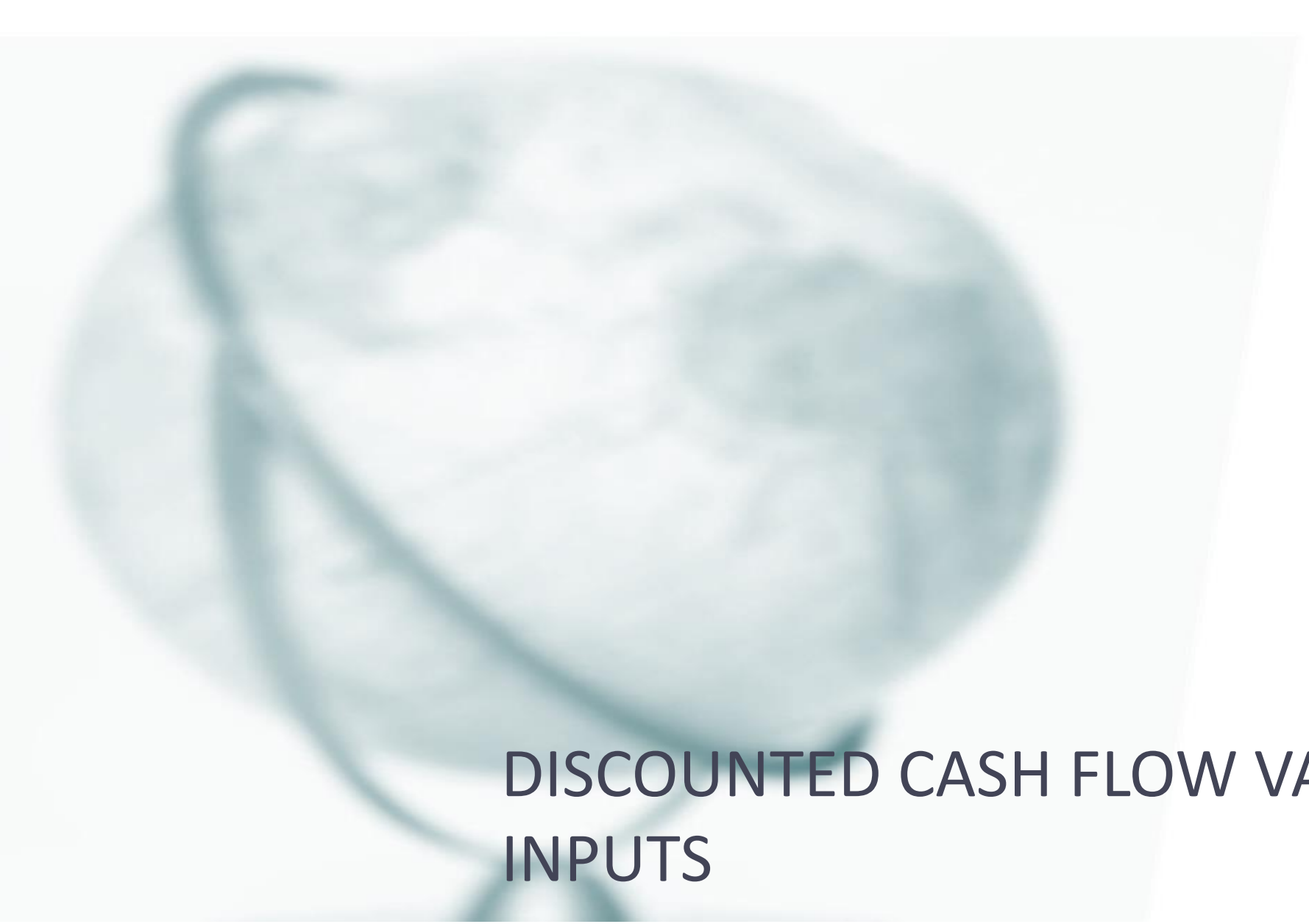
= $ 1873 - $ 800 = **$1073**

First Principle of Valuation

- Discounting Consistency Principle: Never mix and match cash flows and discount rates.
- Mismatching cash flows to discount rates is deadly.
  - Discounting after debt cash flows (equity cash flows) at the weighted average cost of capital will lead to an upwardly biased estimate of the value of equity
  - Discounting pre-debt cashflows (cash flows to the firm) at the cost of equity will yield a downward biased estimate of the value of the firm.

The Effects of Mismatching Cash Flows and Discount Rates

- Error 1: Discount CF to Equity at Cost of Capital to get equity value
  - PV of Equity = $50/1.0994 + 60/1.0994^2 + 68/1.09943 + 76.2/1.0994^4 + (83.49+1603)/1.0994^5 = \$1248$
  - Value of equity is overstated by \$175.
- Error 2: Discount CF to Firm at Cost of Equity to get firm value
  - PV of Firm = $90/1.13625 + 100/1.13625^2 + 108/1.13625^3 + 116.2/1.13625^4 + (123.49+2363)/1.13625^5 = \$1613$
  - PV of Equity = \$1612.86 - \$800 = \$813
  - Value of Equity is understated by \$ 260.
- Error 3: Discount CF to Firm at Cost of Equity, forget to subtract out debt, and get too high a value for equity
  - Value of Equity = \$ 1613
  - Value of Equity is overstated by \$ 540



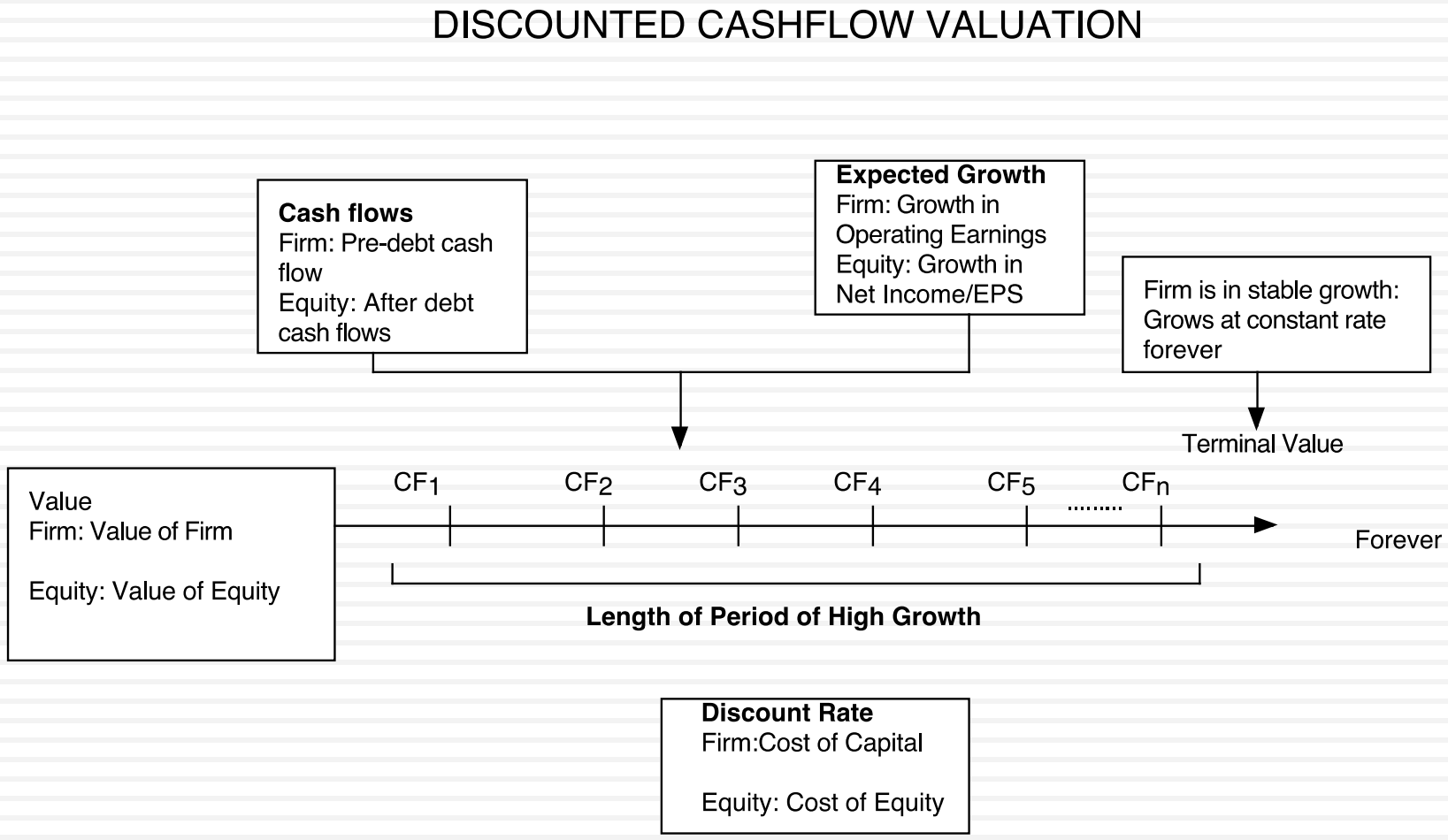
DISCOUNTED CASH FLOW VALUATION: THE INPUTS

The devil is in the details..

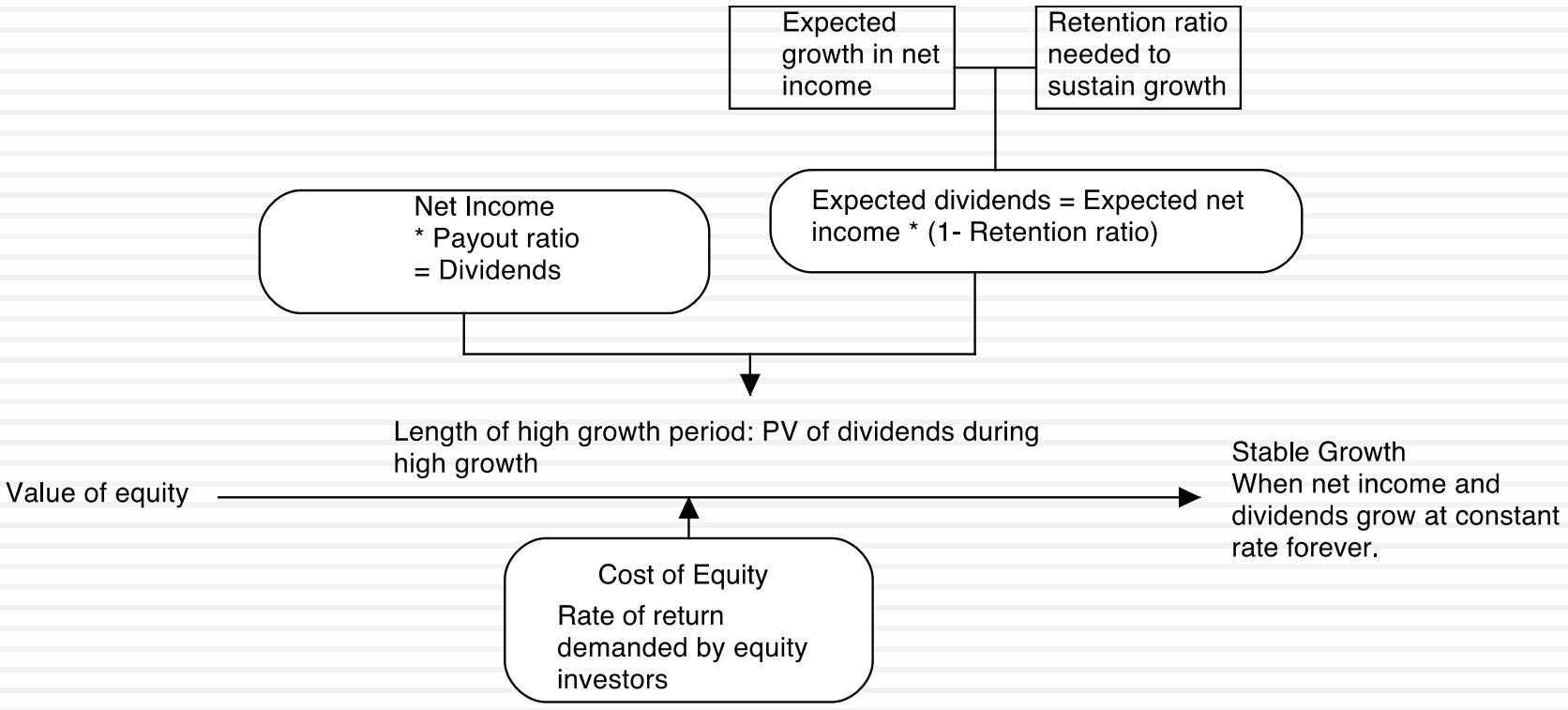
Discounted Cash Flow Valuation: The Steps

1. Estimate the discount rate or rates to use in the valuation
2. Estimate the current earnings and cash flows on the asset, to either equity investors (CF to Equity) or to all claimholders (CF to Firm)
3. Estimate the future earnings and cash flows on the firm being valued, generally by estimating an expected growth rate in earnings. (最难)
4. Estimate when the firm will reach "stable growth" and what characteristics (risk & cash flow) it will have when it does.
5. Choose the right DCF model for this asset and value it.

Generic DCF Valuation Model



Start easy: The Dividend Discount Model



The appendix (1)

- The present value of asset

$$PV = \frac{CF_1}{(1+k)} + \frac{CF_2}{(1+k)^2} + \frac{CF_n}{(1+k)^n}$$

$$= \sum_{t=1}^{n} \frac{CF_t}{(1+k)^t}$$

- If CF is constant, $PV = \frac{CF_1}{k}$
- If CF grows with a rate of g: $CF_{n+1}=CF_n(1+g)$

$$PV = \frac{CF_1}{k-g}$$

The appendix (2)

If CF is constant, $PV = \frac{CF_1}{k}$

If CF grows with a rate of g: $CF_{n+1}=CF_n(1+g)$

$$PV = \frac{CF_1}{k-g}$$

Assume a firm with CF=100, and discount rate k=20%, then

if
- g=0, PV=100/0.20=500
- g=5%, PV=100/0.15=667

→ robust

- g=18%, PV=100/0.02=5000
- g=19%, PV=100/0.01=10000
- g=19.5%, PV=100/0.005=20000

} unstable 不适用

PE ✓ → large firms

VC ✗ → start-up firms

The appendix (3)

- A Two stage model:
  - The cash flow of the asset first grow rapidly for n years
  - Then the cash flow grows with a constant rate g. Then,

$$PV = \sum_{t=1}^{n} \frac{CF_n}{(1+k)^t} + \left(\frac{CF_{n+1}}{k-g}\right) / \frac{1}{(1+k)^n}$$

- The questions:
  - What cash flow?
  - How long does the high growth period last?
  - What discount rate and permanent growth rate to use?